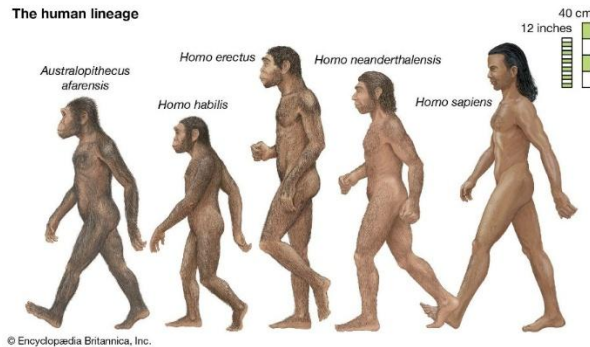


BIOINFORMATICS

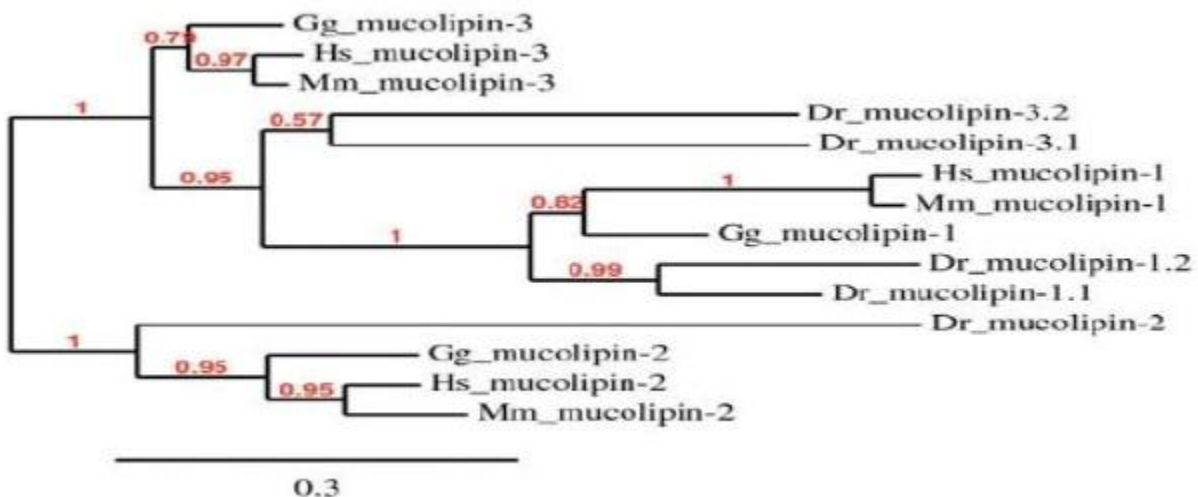
WEEK – 1, DAY – 6

Human Evolution



SIMILARITY VS EVOLUTIONARY RELATIONSHIP

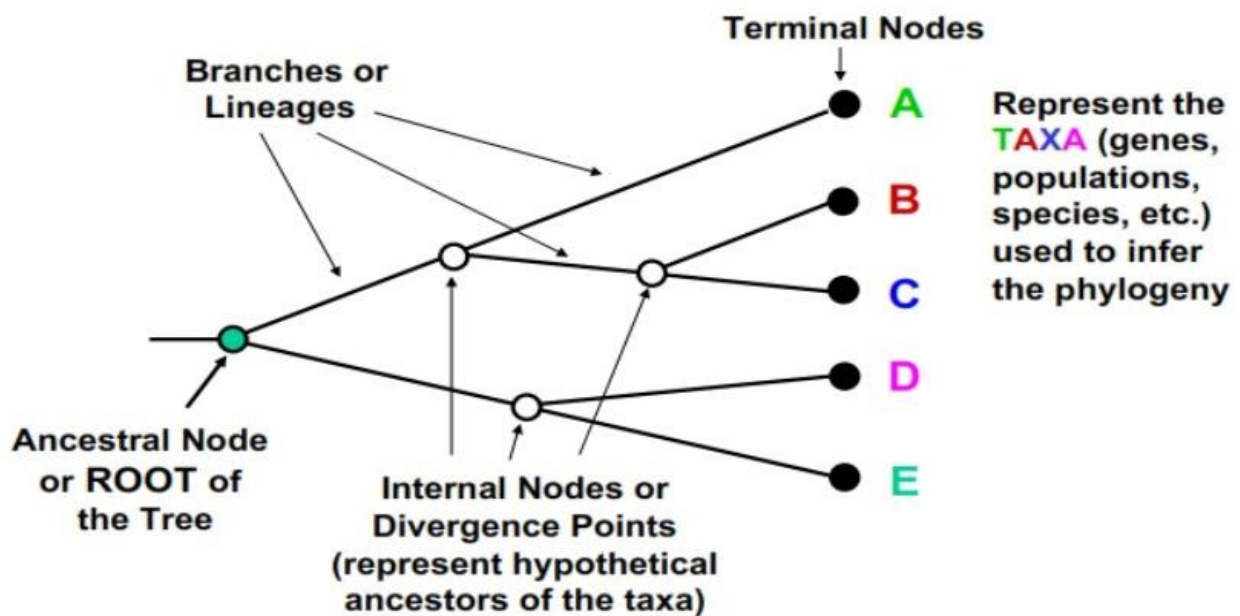
- Similarity and relationship are not the same thing, even though evolutionary relationship is inferred from certain types of similarity.
- **Similar:** having likeness or resemblance (an observation)
- **Related:** genetically connected (an historical fact)
- Two taxa can be most similar without being most closely-related:
 - **Taxonomy** - is the science of classification of organisms.
 - **Phylogeny** - is the evolution of a genetically related group of organisms a study of relationships between collection of "things" (genes, proteins, organs..) that are derived from a common ancestor.
- To Find the evolutionary ties and relationships between organisms or ancestral sequence and its descendants.



A **PHYLOGENETIC TREE** is a branching diagram that represents the evolutionary relationships and history among different species, organisms, or genes based on their genetic or physical characteristics. It acts as an evolutionary family tree, tracking how different organisms diverged from shared ancestors over time

Why Do We Build Phylogenetic Trees?

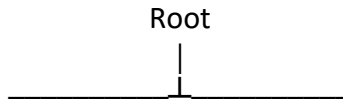
- Study evolution.
- Compare species.
- Investigate disease outbreaks.
- Understand gene evolution.
- Predict functions of unknown genes.
- Trace viral evolution (e.g., influenza, SARS-CoV-2).



The tree has internal nodes, indicating branch points, and external nodes, usually indicating the data points used in constructing the tree. In the context of biological evolution, these data are referred to as Operational Taxonomic Units, or OTUs for short.

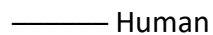
BASIC TERMINOLOGY

1. Root - The **root** represents the common ancestor from which all sequences in the tree evolved.



A rooted tree shows the direction of evolution.

2. Branch - A **branch** represents an evolutionary lineage.



Each branch connects organisms or ancestral nodes.

3. Node - A **node** is where branches split.



The node represents a common ancestor.

4. Leaf (Tip) - The organisms or sequences at the ends of branches are called **tips** or **leaves**.

Examples: Human, Mouse, Cow

5. Clade - A **clade** is a group consisting of a common ancestor and all its descendants.

Example:



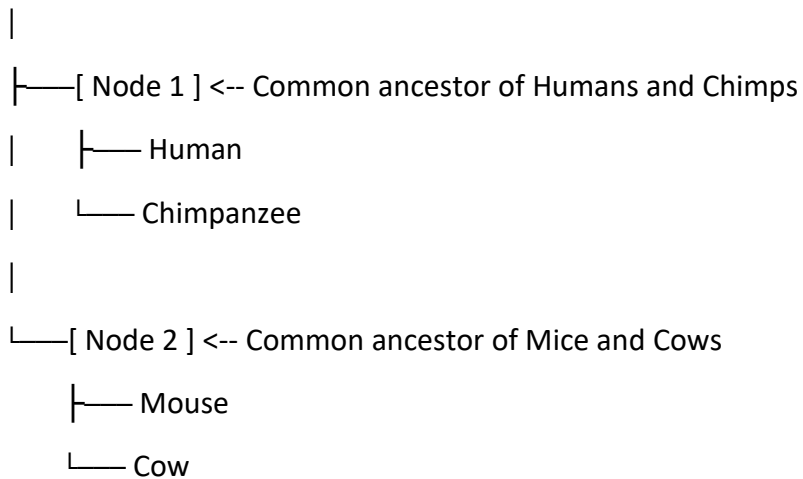
Human and Chimpanzee form a clade because they share a recent common ancestor.

PROPERTIES OF TREES (2)

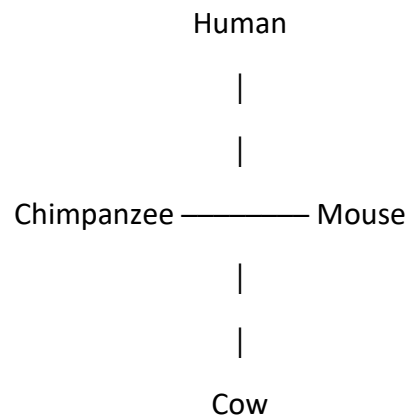
Feature	Rooted Tree	Unrooted Tree
Evolutionary direction	Explicit (from past to present)	None (shows relationships only)
Common ancestor	Clearly identified at the base	Unknown/not represented
Outgroup required	Yes (to anchor the root)	No
Path between two nodes	Single unique chronological path	Shows distance, not direction
Computational complexity	Higher (requires root estimation)	Lower (direct distance calculation)

ROOTED TREE STRUCTURE

[Root] <-- Most recent common ancestor of all taxa



UNROOTED TREE STRUCTURE



BRANCH LENGTH

Not all branches are the same length.

Long branches often indicate:

- More evolutionary changes.
- More sequence differences.

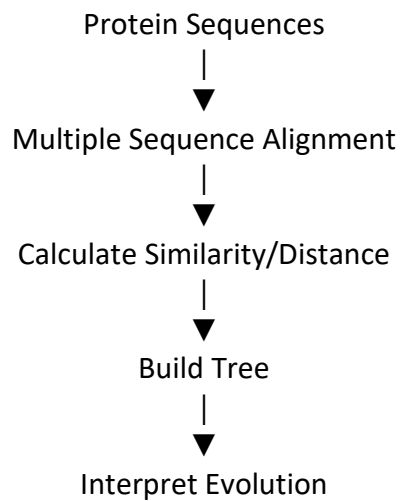
Short branches suggest:

- Fewer changes.
- Closer similarity.

Important: Branch length does **not** necessarily represent time. It often reflects the amount of sequence change.

HOW ARE TREES BUILT?

The basic workflow is:

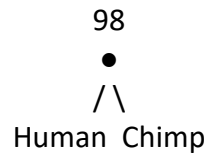


Notice something important: You cannot build a meaningful phylogenetic tree without a good alignment.

BOOTSTRAP VALUES

When software builds a tree, it can estimate how confident it is about each branch. These confidence values are called bootstrap values.

Example:



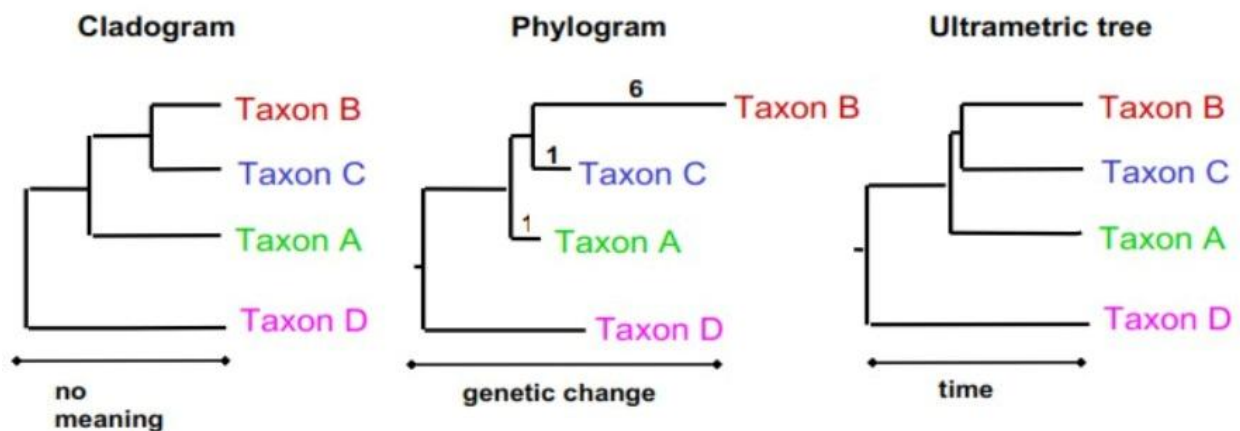
A value of 98 means that, in repeated resampling analyses, this grouping appeared 98% of the time.

General guideline:

Bootstrap	Interpretation
> 90	Very strong support
70–90	Good support
50–70	Weak to moderate support
< 50	Low confidence

higher bootstrap values mean greater confidence.

TYPES OF TREE (3)



1. CLADOGRAM

- **Branch meaning:** No evolutionary meaning.
- **Characteristics:** All the tips align perfectly on the right-hand side. The lengths of the lines are drawn solely to make the diagram neat and readable.
- **Purpose:** Used strictly to show the branching order (topology) and relationships among organisms without displaying mutation rates or time scales.

Applications

- Evolutionary classification
- Teaching phylogeny
- Morphological comparisons

2. PHYLOGRAM

- **Branch meaning:** Proportional to genetic change.
- **Characteristics:** Tips do not align on the right; some lines are much longer than others. The length corresponds to the number of nucleotide or amino acid substitutions per site.
- **Purpose:** Used to show which species or genes are evolving faster or have accumulated more mutations (like the **Clustal Omega** output for your insulin exercise).

Characteristics

- Longer branches indicate greater evolutionary change.
- Branch lengths are meaningful.

Applications

- Molecular evolution
- Mutation rate analysis
- Comparative genomics

3. CHRONOGRAM

- **Branch meaning:** Proportional to chronological time.
- **Characteristics:** The horizontal axis is paired with a time scale calibrated in millions of years (often using fossil records or molecular clocks).
- **Purpose:** Used to determine the exact historical era when two lineages diverged from a common ancestor.

Characteristics

- All present-day taxa terminate at the same time point.
- Often calibrated using fossil records or molecular clocks.

Applications

- Estimating divergence times
- Evolutionary history
- Molecular clock studies